

Do LLMs Understand Language?

LIN 313 Language and Computers

Admin

- HW 4 Due Thursday 11/14
- EC opportunity:
 - coalition of PODER and other groups talking about ongoing campaigns against data centers planned around Austin
 - Sunday: San Marcos Library 2-4pm
- Next week: Instapoll to select topic(s)

Overview

- Chinese Room Argument
 - philosophical objections
- Machine Learning Scaling Laws
- Different ideas of understanding
- and how to test for them in LLMS

John Searle's Chinese Room Argument

Two Hypotheses:

- Weak AI: Computers/programs can effectively simulate intelligent cognition
- Strong AI: A computer can *actually have* a mind. It can literally understand and has genuine cognitive states, not just imitations of them.

The difference isn't about capability or power, but about the **nature of intelligence**.

The Chinese Room Argument is a philosophical argument against the possibility of strong AI.

Let's imagine for a moment the Strong AI hypothesis is true.

John Searle's Chinese Room Argument

There is a room. Sometimes people come to the room with a piece of paper which they slip into the room through a slot. Then they wait a while until the same piece of paper comes back out of a room through a second slot.

The people are native Chinese speakers. They write messages like this.



什麼帶來
快樂

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"What brings happiness?"

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They get back messages like this:



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"Be the stream of the universe."

The answer to the question is highlighted in red. This sentence is from Lao-Tzu's *Tao Te Ching*.

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The people conclude: "There is a fluent Chinese speaker in the room!"

什麼帶來
快樂
爲天下式

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John Searle's Chinese Room Argument

What's actually going on inside the room?

Well, John Searle is in there. He speaks no chinese whatsoever. All he sees is marks on paper.

He has a set of instruction books in the room with him, written in English. Those books tell him how to turn the marks he receives into other marks.

If you see this shape,

"什麼"

followed by this shape,

"帶來"

followed by this shape,

"快樂"

then produce this shape,

"爲天"

followed by this shape,

"下式".



John Searle's Chinese Room Argument

But outside the room,
it looks like there's
someone inside who
speaks Chinese.

video:

https://mind.ilstu.edu/curriculum/searle_chinese_room/searle_chinese_room.html



John Searle's Chinese Room Argument

- Intelligence requires **intentional states**.
 - **intentional states**: the property of mental states being *about something* or directed at something.
- Nothing in the setup has the "right" kind of intentional states
 - Neither Searle nor the room actually understands Chinese—they're just following syntactic rules
 - Searle has intentional states, but not the right kind.
- By analogy, a computer running a program for Chinese is doing the same thing as Searle:
 - it is following formal rules to process symbol strings, but it has **no real understanding** of the language or the conversation.
 - Therefore, running an appropriate program is **not sufficient** for true understanding or a mind. Even if a computer perfectly mimics intelligent behavior, it doesn't necessarily have genuine understanding or consciousness.
- Conclusion: Strong AI is impossible

Essential argument: **Syntax** (symbol manipulation) doesn't equal **semantics** (meaning and understanding).

Philosophical objections to the CRA

1. The Systems Reply

- a. While the individual person in the room doesn't understand Chinese, the entire system (person + rulebook + room) does understand Chinese, just as individual neurons don't understand language but the brain as a whole does.

2. The Robot Reply

- a. A computer connected to sensors and actuators that interacts with the real world (embodied AI) would have genuine understanding because it could connect symbols to real-world referents, unlike the isolated room.

3. Other Minds Problem

- a. We can't truly know if anyone else understands—we only observe behavior. If we accept that other humans understand based on their behavior, why not accept that a computer passing the same behavioral tests understands?

4. Begging the Question

- a. Searle assumes from the start that following rules mechanically can't produce understanding, which is precisely what's being debated. The thought experiment may simply reflect our intuitions rather than proving anything about what understanding requires.

5. There are more...

If it walks like a duck and talks like a duck...

The results are in and it seems like they don't respect philosophy.

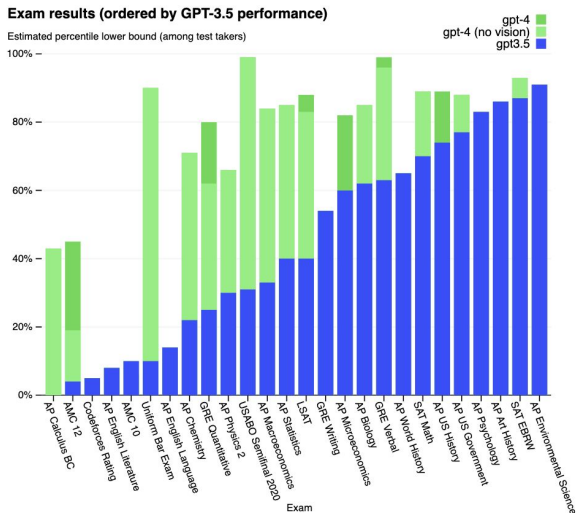


Figure 4. GPT performance on academic and professional exams. In each case, we simulate the conditions and scoring of the real exam. Exams are ordered from low to high based on GPT-3.5 performance. GPT-4 outperforms GPT-3.5 on most exams tested. To be conservative we report the lower end of the range of percentiles, but this creates some artifacts on the AP exams which have very wide scoring bins. For example although GPT-4 attains the highest possible score on AP Biology (5/5), this is only shown in the plot as 85th percentile because 15 percent of test-takers achieve that score.

	GPT-4 Evaluated few-shot	GPT-3.5 Evaluated few-shot	LM SOTA Best external LM evaluated few-shot	SOTA Best external model (incl. benchmark-specific tuning)
MMLU [49] Multiple-choice questions in 57 subjects (professional & academic)	86.4% 5-shot	70.0% 5-shot	70.7% 5-shot U-PaLM [50]	75.2% 5-shot Flan-PaLM [51]
HellaSwag [52] Commonsense reasoning around everyday events	95.3% 10-shot	85.5% 10-shot	84.2% LLaMA (validation set) [28]	85.6 ALUM [53]
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DROP [58] (F1 score) Reading comprehension & arithmetic.	80.9 3-shot	64.1 3-shot	70.8 1-shot PaLM [3]	88.4 QDGAT [59]
GSM-8K [60] Grade-school mathematics questions	92.0% * 5-shot chain-of-thought	57.1% 5-shot	58.8% 8-shot Minerva [61]	87.3% Chinchilla + SFT+ORM-RL, ORM reranking [62]

Table 2. Performance of GPT-4 on academic benchmarks. We compare GPT-4 alongside the best SOTA (with benchmark-specific training) and the best SOTA for an LM evaluated few-shot. GPT-4 outperforms existing LMs on all benchmarks, and beats SOTA with benchmark-specific training on all datasets except DROP. For each task we report GPT-4’s performance along with the few-shot method used to evaluate. For GSM-8K, we included part of the training set in the GPT-4 pre-training mix (see Appendix E), and we use chain-of-thought prompting [11] when evaluating. For multiple-choice questions, we present all answers (ABCD) to the model and ask it to choose the letter of the answer, similarly to how a human would solve such a problem.

Ray Kurzweil

1 The accelerating pace of change ...



2 ... and exponential growth in computing power ...

Computer technology, shown here climbing dramatically by powers of 10, is now progressing more each hour than it did in its entire first 90 years

COMPUTER RANKINGS

By calculations per second per \$1,000



Analytical engine
Never fully built, Charles Babbage's invention was designed to solve computational and logical problems



Colossus
The electronic computer, with 1,500 vacuum tubes, helped the British crack German codes during WW II



UNIVAC I
The first commercially marketed computer, used to tabulate the U.S. Census, occupied 943 cu. ft.



Apple II
At a price of \$1,298, the compact machine was one of the first massively popular personal computers



Power Mac G4
The first personal computer to deliver more than 1 billion floating-point operations per second

3 ... will lead to the Singularity

2045
Surpasses brainpower equivalent to that of all human brains combined

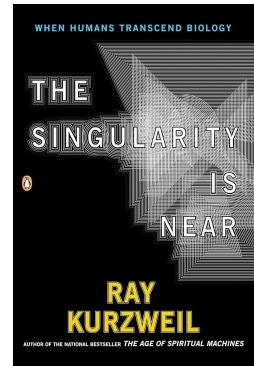
Surpasses brainpower of human in 2023



Surpasses brainpower of mouse in 2015



Source: Time Magazine



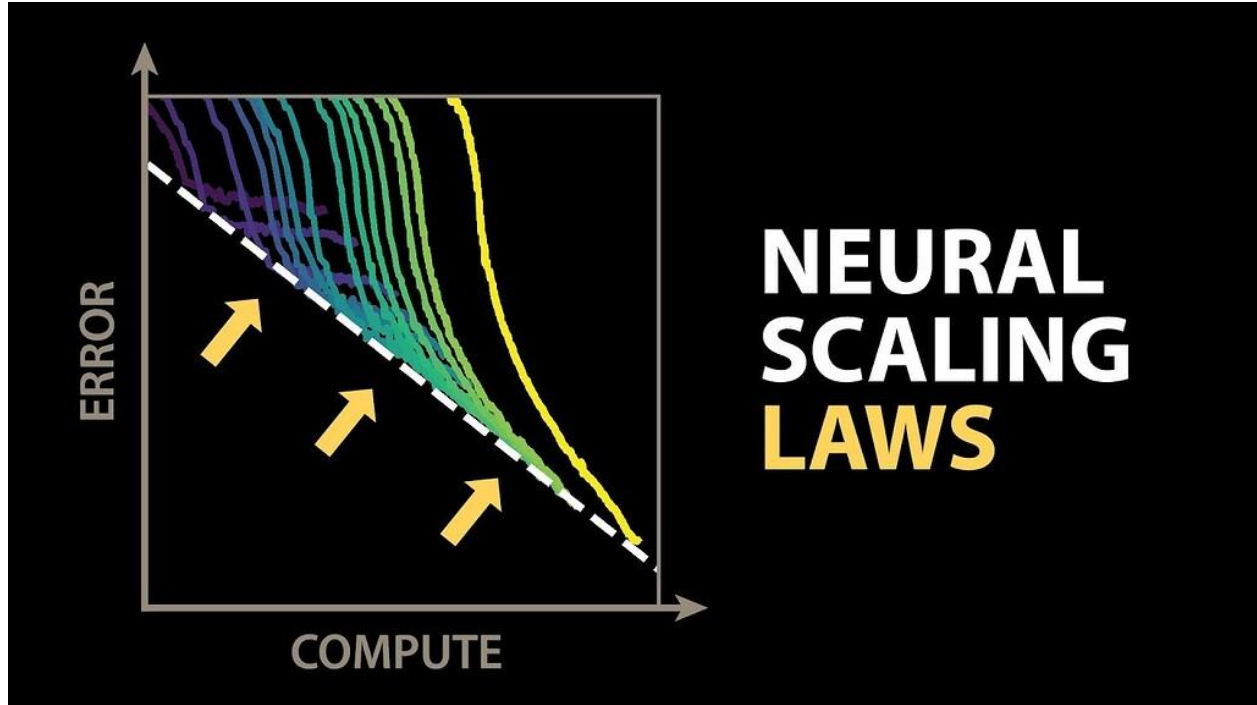
Scaling LLMs: How big can we go?

GPT-5 optimized its performance almost exactly as predicted for this combination of three factors.

Model size (N) -
number of parameters

Dataset size (D) -
amount of training data

Compute (C) - amount
of computation used for
training



<https://www.youtube.com/watch?v=5eqRuVp65eY&t=546s>

Emily Bender's Octopus

Essentially the CRA for LLMs.

1. A and B are stranded on desert islands. They communicate through an undersea cable.
2. An octopus underwater intercepting messages between A and B, and learns how to imitate B.
3. EMERGENCY!

Argument:

- The octopus learns pure statistics, and no amount of statistics about *form* can ever fully represent *meaning*
- Meaning requires *grounding*—connection to the world, to perception, action, and communicative intent

Her point is social:

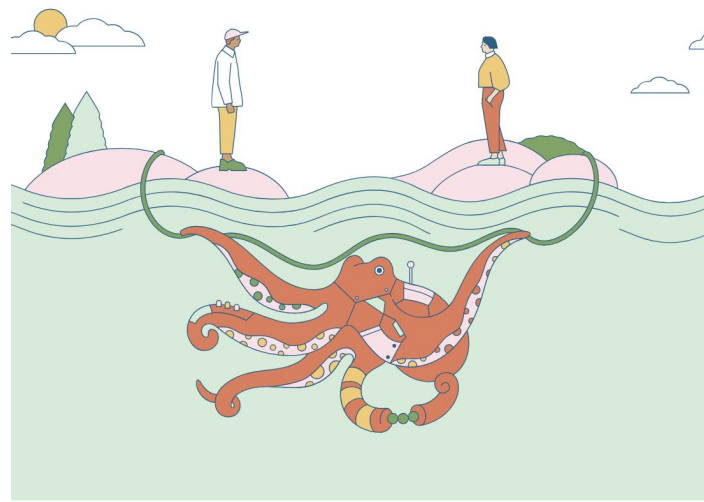
- LLMs are dangerous. They produce harmful or misleading content because they lack understanding
- They can't reliably distinguish truth from plausible-sounding falsehoods
- Their apparent competence can be deceptive



Climbing towards NLU: On Meaning, Form, and Understanding in the Age of Data

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Who cares if it's strong as long as it's general

Strong AI is a high bar—maybe too high, for most purposes.

A computer that can effectively simulate human cognition would be very interesting in its own right.

These days people prefer the terms **AGI** (artificial general intelligence) and **superintelligence**.

Strong AI: actually has a mind

- sentience, a 'soul', personhood, etc....

AGI: walks like a duck, talks like a duck

- passes our strongest version of the turing test

What is Understanding? (For Mitchell)

Three domains of understanding

- intuitive physics
- intuitive psychology
- intuitive biology

Two cognitive-scientific ideas of understanding

- **Simulation: predicting possible futures**
 - Barsalou (1999): situated concepts: Understanding and reasoning involve mentally simulating relevant perceptual, motor, and introspective experiences
- **Abstraction and analogy**
 - Doug Hofstadter (2007): "the perception of a common essence between two things"
 - Bruno Latour (1987): abstractions are not immaterial, but material accomplishments that require maintenance and support. Abstract concepts don't exist "above" or separate from the material world; they're held in place by concrete practices and connections.



LLMs Simulate the (entire) future

Transformers Represent Belief State Geometry in their Residual Stream

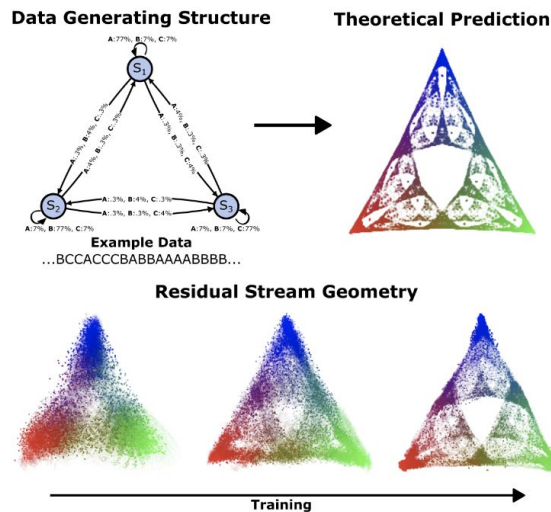
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Pitzer and Scripps College

Lucas Teixeira
PIBBSS

Alexander Gietelink Oldenzien
University College London
Timaeus

Paul M. Riechers
Simplex
BITS



- trained small transformer models on artificially generated "data" from a stochastic markov model
- trained model to predict just the next token
- demonstrated that the model keep track of different possibilities for what the future might hold

LLMs have reason???

Exam results (ordered by GPT-3.5 performance)

Estimated percentile lower bound (among test takers)

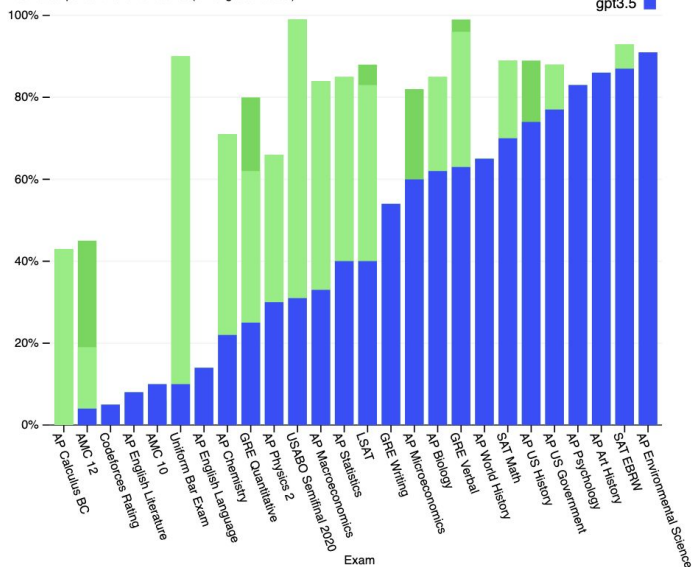


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LLMs Have Reason?

GPT's test results look pretty good! If it can do PhD level math and chemistry problems, it must have abstract reasoning abilities, right?

WRONG

data contamination: overlap between training data and test data.

ChatGPT 4o 

 Share

RT

There is a one-story house where everything is pink.
The windows are pink, the floors are pink, the walls are
pink - everything is pink! What color are the stairs?

LLMs appear to solve tricky brain teasers

ChatGPT 4o ▾

↑ Share

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There are no stairs in a one-story house!

But they've seen a lot of tricky brain teasers!

ChatGPT 4o ▾

↑ Share

RT

There is a two-story house where everything is pink.
The windows are pink, the floors are pink, the walls are
pink - everything is pink! What color are the stairs?

But they've seen a lot of tricky brain teasers!

ChatGPT 4o ▾

↑ Share

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The house doesn't have stairs because it is a **two-story house**, which implies there might not be stairs at all.

Testing for Abstraction (Thomas McCoy et al.)

Q: How are we supposed to know if the model actually learned an abstract rule or just learned shallow heuristics like memorizing a bunch of examples (a.k.a. spurious correlations)??

A: Look at how it generalizes on out-of-domain (Ood) questions

PNAS

RESEARCH ARTICLE

PSYCHOLOGICAL AND COGNITIVE SCIENCES
COMPUTER SCIENCES

OPEN ACCESS



Embers of autoregression show how large language models are shaped by the problem they are trained to solve

R. Thomas McCoy^{A1,2,3}, Shunyu Yao^{A4}, Dan Friedman^A, Mathew D. Hardy^B, and Thomas L. Griffiths^{A,B}

Deciphering the Factors Influencing the Efficacy of Chain-of-Thought: Probability, Memorization, and Noisy Reasoning

Akshara Prabhakar¹, Thomas L. Griffiths^{1,2}, R. Thomas McCoy^{3,4}

Article Swapping

Task: Swap each article (a, an, the) with the previous word

Article swapping

Swap each article (*a*, *an*, or *the*) with the word before it.

Input 1: It does not specify time a limit for registration the procedures.

Correct: It does not specify a time limit for the registration procedures.

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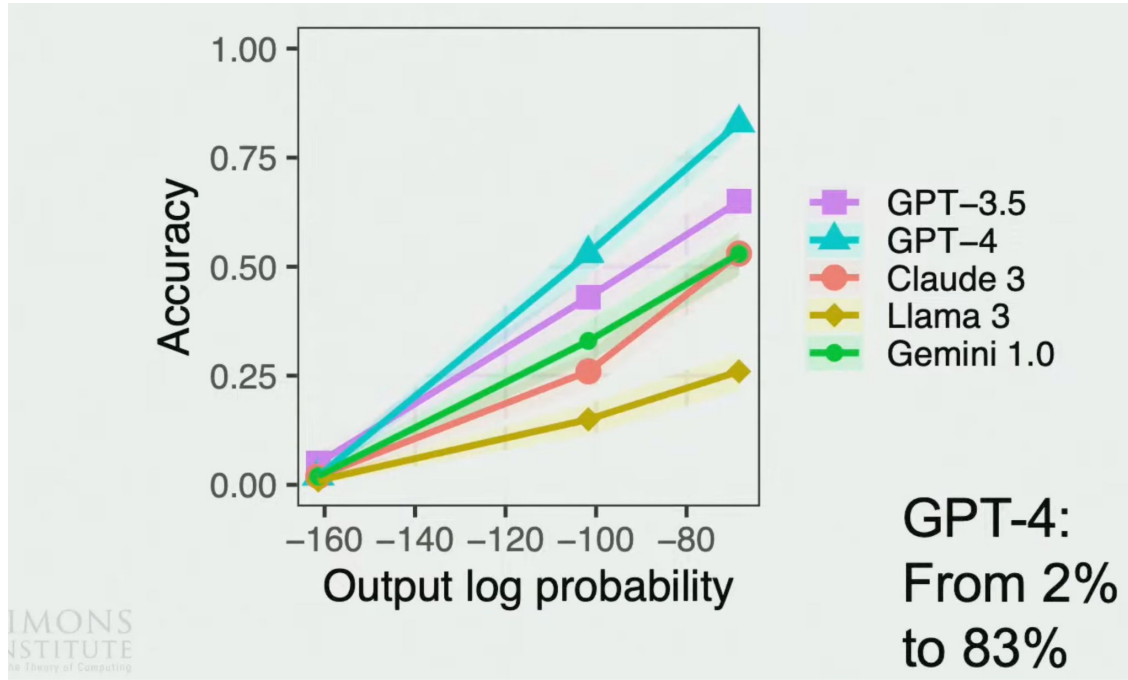
Input 2: It few with it to lying take the get just a hands would kinds.

Correct: It few with it to lying the take get a just hands would kinds.

✗ **GPT-4:** It flew with a few kinds to take the lying just to get the hands.

Article Swapping: Probability to Sensitivity

The model does a lot better at the task if the input string is high probability (aka a grammatical sentence)



Counting words

- The task:
 - How many words are in this list? "lively news exhibit sleep"

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